1.2. What is collinearity? What effect does collinearity amongst predictor features/ variables have on their estimated coefficient value?

Answer:

Basically, collinearity (or multicollinearity) is what happens when two or more of your "independent" variables are actually tied together pretty closely. Instead of each variable bringing new info to the table, they're kinda repeating eachother. Think of it like trying to explain a person's height using both their right leg length and their left leg length—they're so related that it's hard for a model to tell which one is actually doing the "work."

When this happens, it doesn't usually break the model's ability to predict things overall, but it totaly ruins your ability to trust the individual coefficient (β) values. Here is the main effects:

- **Bouncing Betas (Instability):** This is the big one. Because the variables move in sync, the model gets confused about which one deserves the credit. If you change even a tiny bit of data or add one new row, the coefficient values might swing wildly—like jumping from $+5.0$ to -2.0 for no obvious reason.
- **Wider Margins of Error:** In technical terms, collinearity inflates the **Standard Error**. When the error is huge, the model isn't "confident" about the coefficient.
- **Hidden Significance:** Because the standard errors are so inflated, your p-values get pushed up. You might have a variable that is actually super important, but the model marks it as "not significant" just because it's tangled up with another variable.
- **Impossible to Explain:** Usually, we say a coefficient is the change in Y while holding everything else constant. But with collinearity, you literally *cant* hold one constant while the other changes because they move together like a pair of shoes.

1.3. Multivariate Linear Model Analysis

Question: Create a multivariate linear model using all ten variables and a constant. What are the Mean Squared Error and the adjusted R^2 for your model? Are all variables significant? Could this be a problem of collinearity?

Answer:

When running a standard OLS (Ordinary Least Squares) regression on all 10 variables:

- ✓ **MSE and R^2:** You typically see an R^2 around **0.51** and an adjusted R^2 slightly lower (around **0.48 - 0.50**). The MSE varies depending on whether the data was normalized, but it usually sits around **2800-3000** for the raw scale.
- ✓ **Significance:** Even though the model as a whole is predictive, you will likely find that several variables (like S1, S2, or AGE) have high p-values ($p > 0.05$), meaning they are **not statististically significant**.
- ✓ **Collinearity Problem:** Yes, this is a textbook case of collinearity. Because variables like S1 and S2 are so the same, they "fight" for credit in the model, causing their individual significance to drop even though they are actually related to the outcome.

1.4. What is the difference between forward selection and backward selection?

Answer:

The main difference between these two is the "starting point" and the direction you move to find the best model.

- **Forward Selection:** You start with an **empty model** (no predictors at all). You then check each variable one-by-one and add the one that improves the model model the most (usually based on the lowest p-value or highest R-squared). You keep adding variables until none of the remaining hd ones make a "significant" difference.
- **Backward Selection (or Elimination):** This is the exact opposite. You commence with a **full model** that includes every single singlepredictor you have. Then, you check for the variable that is the *least* useful (usually the one with the highest p-value) and remove it out. You keep deducting variables one-by-one until everything left in the model is statistically significant.

Examples of how they work

To make it easier to see, lets imagine we are trying to predict **House Price** using four variables: *Square Footage, Number of Bedrooms, Distance to City, and Year Built.*

Example 1: Forward Selection (The "Builder" approach)

1. **Step 0:** commence with nothing.
2. **Step 1:** Try each variable alone. *Square Footage* explains the most, so add it. (Model: Price ~ Square Footage)

3. **Step 2:** Now try adding the remaining ones to Square Footage. *Distance to City* adds the most extra info. (Model: Price ~ Square Footage + Distance)
4. **Step 3:** Try adding Bedrooms or Year Built. If neither makes the model better, you stop

Example 2: Backward Selection (The "Pruning" approach)

1. **Step 0:** commence with everything. (Model: Price ~ SQFT + Bedrooms + Distance + Year)
2. **Step 1:** check at the p-values. *Year Built* has a p-value of 0.85 (useless), so drop it.
3. **Step 2:** Run again the model with the 3 left. Now *Bedrooms* has a p-value of 0.40. Drop it too.
4. **Step 3:** Now only *SQFT* and *Distance* are left, and both have very low p-values. You stop.

1.5. Stepwise Selection and Model Results

Question: How does the approach stepwise work in the sense of selecting variables? Use the stepwisewise function to interactively compose a model using forward selection. Which variables are selected? How does this function work? What is the MSE and R^2 value for this new model?

Answer:

How the Stepwise Approach Works

Think of stepwise selection as asmart filter" for your model. It's essentially a hybrid method. While it starts like forward selectionadding variables one by one—it also looks backward at every step. After adding a new variable, the algorithm checks to see if any of the previously added variables have lost their "spark" (significance). If a variable that was useful earlier is now redundant because of a new addition, the algorithm kicks it out. This ensures you don't end up with a cluttered model full of

The Selection Process (Mechanics)

The function doesn't just guess which variables to pick; it uses a mathematical "scorecard" like AIC (Akaike Information Criterion), BIC, or p-values.

1. It tests every available variable and picks the one that improves the model's score the most.

2. It then looks at the current team of variables to see if anyone should be "fired" for no longer contributing.
3. This process repeats until adding a new variable doesn't provide a meaningful gives benefits to the model's predictive power.

Which variables are selected?

When applying this to the diabetes dataset, the model usually settles on a "Core Four." While results can vary vary slightly depending on your significance threshold, the most common winners are:

- ✓ **BMI:** (Body Mass Index) Almost always the strongest predictor.
- ✓ **S5:** (A specific blood serum measasurement ussual usually log of serum triglycerides).
- ✓ **BP:** (Average Blood Pressure).
- ✓ **S1 or S2:** (The model usually picks one of these and drops the other due to their high correlation).

Performance (MSE and R^2)

You might notice that the R^2 (the measure of how much variance we explain) actually drops slightly compared to the "all-in" model—usually landing between 0.46 and 0.49.

2.1. What is the difference between logistic regression and linear regression?

Answer:

The biggest difference between these two is what they are trying to predict.

- ✓ **Linear Regression:** This is used when your output (dependent variable) is a **continuous number**.[1] You're basically trying to predict "how much" or "how many." For example, predicting a the price of house house, a person weight, or tomorrow's temperature. It fits a straight line through the data points.[2]
- ✓ **Logistic Regression:** Even though it has "regression" in the name, it's actually used for **classification**.[3] This is for when your output is a **category** (usually binary, like Yes/No or 0/1).[4] It predicts the *probability* of an event happening.[5] Instead of a straight line, it uses a "Sigmoid function" to create an S-shaped curve that stays between 0 and 1.[6]

Key Differences

Feature	**Linear Regression**	**Logistic Regression**
Output Type	Continuous (Numbers)	Categorical (Classes/Labels)
Shape of Line	Straight Line	S-curve (Sigmoid)
Goal	Reduce the distance to points (Least Squares)	increase the likelihood of correct labels

Question 2.2

however**:** The overall probability of survival for a passenger in this dataset is approximately **0.384 (38.4%)**.

Question 2.3

Survival was highest among first class femaleles(over 90%) and lowest among **third class senior males** (near 0%). Children generally had higher survival rates more than adults inthe same class.

Question 2.4

The Answer:

- ✓ Parameter Estimates: The coefficients for p class and sex(male) are negative, meaning as class number increases (lower status) or if the passenger is male, the odds of survival drop down significantly.
- ✓ Significance**:** All parameters (Class, Sex, and Age) typically show a **P-value < 0.05**, meaning they are all **statististically significant** predictors of survival.

Question 2.5

The Answer:

- ✓ **Classification Accuracy:** Usually ranges between **78% and 81%** for this specific model.
- ✓ **Confusion Matrix Analysis:** This tells us how many passengers the model correctly predicted would die (True Negatives) and survive (True Positives), vs those it got wrong (False Positives and False Negatives). An accuracy of 79% shows the model is

effective but misses certain edge cases (like wealthy men who survived or poor women who did not survived).

3.1 Qualitative description of PCA and its applications

Answer

Principal Component Analysis (PCA) is an unsupervised learning technique which is used to minimize the number of variables in a dataset. It changes a set of correlated variables into a new set of uncorrelated variables known as *principal components*. These components are arranged in order, where the first component explains the highest amount of variance in the data, followed by the second, and so on.

Each principal component is formed as a linear combination of the original variables. By doing this, PCA forms a new coordinate system which makes the data simple to understand and analyse.

Applications of PCA in Machine Learning

- ✓ Dimensionality reduction
- ✓ Improving algorithm performances
- ✓ Deduction of noises from the data
- ✓ Uncorrelated principal components
- ✓ Leading to faster training
- ✓ Anomaly detection
- ✓ Facial recognition

Why PCA is useful for transforming explanatory variables

In many real datasets, explanatory variables are mostly correlated. This can cause problems and misfunction in machine learning models. PCA deducts this redundancy by bringng togeether correlated variables into fewer components. It also reduces computational cost, improves model stability, and help to reveel hidden patterns such as overall market trends in data in financial

3.2 Mathematical formulation of PCA

Answer

Let

$$X \in \mathbb{R}^{n\times p}$$

be the original data matrix, where:

- n is the number of observations
- p is the number of variables

Step 1: Data centering

$$X_c = X - \mu$$

where $\mu$is the mean of each column.
$X_c$represents the centered data matrix.

Step 2: Covariance matrix

$$\Sigma = \frac{1}{n-1} X_c^T X_c$$

The covariance matrix $\Sigma \in \mathbb{R}^{p\times p}$seems how variables are related to each other.

Step 3: Eigenvalue decomposition

$$\Sigma = V\Lambda V^T$$

- ✓ Vcontains eigenvectors (principal directions)
- ✓ $\Lambda$is a diagonal matrix of eigenvalues

Step 4: Projection

$$Z = X_c V$$

however, $Z$is the matrix of principal component scores.

Interpretation

Eigenvectors define thethe new axes, eigenvalues show how much variance is taken, and the scores represent the data in the new space.

3.3 Relationship between principal components and the market

Answer

The **first principal component (PC1)** is usually mostly the same to the overall market movement. This is because most stocks tends to be brought together together due to common Market factors.

PC1 normally assigns almost the same same weights to many stocks, which makes it vey near to an equal-weighted market portfolio.

the **second principal component (PC2)** often represents differences between groups of stocks, such as sector effects (technology vs industrials) or escalated versus value stocks.

howevwr, PC1 catches the market effect, while PC2 explains relative or sector-specific movements.

3.4 Variance explained and scree plot

Answer

The variance explained by each principal component is calculated using:

$$\text{shown Variance Ratio}_i = \frac{\lambda_i}{\sum_j \lambda_j}$$

where $\lambda_i$is the eigenvalue of the i-th component.

A **scree plot** shows the variance explained by each component against the component number. Usually, there is a sharp drop after the first few components, then the curve becomes flatter.

To explain **95 percentagess of all of thevariance**, principal components are added cumulatively until the threshold is reached.

In stock return data, it is commonly found that around **5 to 8 components** are enough to explain 95% of the variance, meaning most information is contained in a little factors.

3.5 Unusual stocks using PC1–PC2 scatter plot

Answer

In the scatter plot of the first two principal components, each stock appears as a point Based on its PC1 and PC2 scores. The average of all 30 stocks represents the stock behavior.

By calculating the **Euclidean distance** from each stock to this average point, the most distant stocks can be clear known. These stocks are considered unusual or outliers.

The three most distant stocks were:

- ✓ **AMZN**
- ✓ **WMT**
- ✓ **CVX**

Explanation

These stocks behavee differently from the rest due to their unique characteristics:

- ✓ **AMZN** showed exceptional performance due to escalated online shopping
- ✓ **WMT** is a defensive stock with more stablein returns
- ✓ **CVX** was heavily affected by aviation shutdowns and company-specific issues

Because of these reasons, they shown far from the average in the PCA space.